AI-Driven Stock Screener

(NSE 500 News Filter)

~100 Interview Questions

For candidates & interviewers exploring this project

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1. Project Overview & Architecture

1. What is the primary purpose of this project, and what problem does it solve?
2. Walk me through the end-to-end data flow from stock list to final ranked output.
3. What are the main Python modules and their responsibilities in this codebase?
4. Describe the overall architecture. Is it layered? How are the layers separated?
5. Why did you choose Streamlit for the UI layer? What are its strengths and limitations?
6. How does the concept of a 'universe' of 500 stocks work, and why is it important?
7. What would happen if one module (e.g. news_fetcher) fails? Does the whole app crash?
8. How does the app decide between ML-based scoring and a deterministic fallback?
9. What design patterns or principles (e.g. separation of concerns, OOP) are used here?
10. If you were to redesign this system from scratch, what would you do differently?

2. Data Fetching & Processing

11. How does DataFetcher load the stock universe, and what happens if the CSV is missing?
12. What data does the yfinance library return, and how is it normalized/standardized?
13. Why does fetch_historical_ohlcv reset the DataFrame index? What columns must be present?
14. What are the default fundamental metrics, and why are they used as safe fallbacks?
15. How are missing or invalid fundamental values handled?
16. What is the role of the 'period' parameter, and how does it affect the data fetched?
17. How would you handle API rate limiting when fetching data for 500 tickers?
18. What potential data quality issues exist with yfinance data, and how would you mitigate them?
19. Why is a sequential fetch loop potentially a bottleneck? How could you parallelize it?
20. How would you cache fetched data to disk to avoid re-hitting the API on every run?

3. Feature Engineering & Technical Indicators

21. What technical indicators does IndicatorEngine compute, and why these specific ones?
22. Explain how RSI is calculated and what it signals. What does an RSI of 50 mean?
23. What is the MACD, and what do its three components (MACD, Signal, Histogram) represent?
24. How are Bollinger Bands constructed, and how is the BB_Position feature derived?
25. What does the SMA_Ratio feature represent, and how is it a 'trend' signal?
26. How is the Volume_Trend feature calculated, and why use a 5-day vs 20-day average?
27. What happens in compute_technical_indicators when there are fewer than 200 rows?
28. Why are default values like RSI=50 and MACD=0 injected for short histories?
29. How does extract_current_features build the final feature vector from the last row?
30. How is look-ahead bias avoided when computing features at each point in time?
31. Which features are technical, which are fundamental, and which are news-based?
32. How would you handle division by zero when computing BB_Position or SMA_Ratio?

4. Machine Learning & Model Training

33. What type of model is used, and why was RandomForestClassifier chosen?
34. How is the training data generated? Explain the walk-forward labeling approach.
35. What is the label in this supervised learning problem, and how is it derived?
36. How does forward_days affect the label definition? What trade-offs exist?
37. What is the purpose of the sample_stride parameter, and what problem does it solve?
38. Why does the code skip the first 200 rows during training?
39. How is the train/test split performed, and why use stratification?
40. What is class imbalance, and how does class_weight='balanced' address it?
41. What metrics are reported (accuracy, ROC-AUC)? Why might accuracy be misleading here?
42. Why is the model refit on the full dataset after evaluation? Is that a good practice?
43. What hyperparameters are used (n_estimators, max_depth, min_samples_leaf)? Why?
44. How does the feature schema stay consistent between training and inference?
45. What is the risk of news features being all-zero during training? How is it handled?
46. How does AIScorer convert model probabilities into a 0-100 score?
47. What happens if predict_proba fails? What fallback score is returned?
48. How would you evaluate this model's real-world predictive value? What are its limits?

5. News & Sentiment Analysis

49. Where does the news data come from, and what API key is required?
50. How does NewsFetcher guarantee it never crashes the main pipeline?
51. What happens if no API key is set? What feature values are returned?
52. Explain the news sentiment normalization logic (handling 0-100 vs -1 to 1 scales).
53. How is recency weighting computed from the published_at timestamp?
54. What is the difference between News_Sentiment_Avg and News_Sentiment_Weighted?
55. How is News_Count_24h calculated, and why limit to 24 hours?
56. What are the five news features, and how might they influence the final score?
57. Why does the model_trainer use neutral defaults for news rather than real data?
58. What are the risks of using news sentiment in a trading signal?

6. Streamlit UI, Caching & Performance

59. How does st.cache_data improve performance, and what is the TTL set to?
60. What is the purpose of the force_refresh checkbox, and how does it clear the cache?
61. How does the app limit the number of tickers processed, and why?
62. Explain the use of st.progress for the pipeline. How is its value clamped?
63. How are the sidebar filters (sector, AI score, P/E, growth) applied to the DataFrame?
64. How does the market cap category filter (Large/Mid/Small) work?
65. What Plotly visualizations are rendered, and what do SMA 50/200 overlays show?
66. Why does the chart section fall back from '6m' to '1y' period data?
67. How is the CSV export functionality implemented? What encoding is used?
68. How would you optimize the app if it takes too long to process 500 tickers?

7. Error Handling & Edge Cases

69. List the key places where exceptions are caught and gracefully handled.
70. What happens if fetch_historical_ohlcv returns fewer than 20 rows in app.py?
71. How does the safe_float helper prevent crashes when converting fundamentals?
72. What happens if no stocks pass the filters (empty display DataFrame)?
73. How does the app handle a ticker whose chart data returns empty for both 6m and 1y?
74. What default values are used when fundamentals are missing, and are they sensible?
75. How does IndicatorEngine handle empty DataFrames in extract_current_features?
76. What happens in model_trainer if no training data can be collected?
77. How are NaN values in indicator calculations (e.g. RSI) handled?
78. What edge cases exist in the fallback scoring formula (e.g. negative PE)?

8. Testing & Code Quality

79. What testing framework is used, and how do you run the test suite?
80. What does test_load_universe_structure verify, and why does it use pytest.skip?
81. How does test_technical_indicators_flat_input simulate a flat market?
82. Explain the no-lookahead test in test_model_trainer.py and why it matters.
83. How does the fake OHLCV generator avoid hitting the network during tests?
84. What boundary conditions does test_scorer_boundary_constraints check?
85. What testing gaps exist in this project? What would you add?
86. How could you use mocking to test fetch_fundamental_metrics without a live API?

9. Deployment, Tooling & Productionization

87. How is the app launched, and what port does it use in launch_streamlit.py?
88. What is the purpose of the health check loop in launch_streamlit.py?
89. Explain the role of the .bat file and the launcher logs.
90. How would you containerize this app with Docker? What would the Dockerfile contain?
91. How would you deploy this to the cloud (e.g. Streamlit Cloud, AWS, Heroku)?
92. How would you manage the MARKETAUX_API_KEY securely in production?
93. What CI/CD steps would you add for this project?
94. How would you handle secrets and environment-specific configuration?

10. Financial Domain & Design Decisions

95. Why is a disclaimer ('not financial advice') included? What are the ethics here?

96. What does the 'Top 50' ranking actually represent, and how is it derived?

97. How would you explain the difference between technical and fundamental analysis as used here?

98. What are the main limitations of using daily historical data + indicators for prediction?

99. How could market regime changes (bull vs bear) affect model performance?

100. If you were to add a new feature (e.g. options data, macro indicators), how would you?

101. How would you backtest this scoring strategy and measure its risk-adjusted returns?

102. What ethical and regulatory concerns arise from an AI stock-picking tool?